

A05 — Chat GPT

Research · Interaction · Reflection on Machine Learning and Generative AI

Student	Team	Course	Due Date
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Part 1 — Understanding ChatGPT

How Does ChatGPT Learn?

ChatGPT learns through a three-stage training process that combines massive-scale unsupervised learning, supervised fine-tuning, and reinforcement learning from human feedback (RLHF). Each stage builds on the previous, transforming a raw statistical model into a conversational assistant capable of nuanced, contextually appropriate responses (IntuitionLabs, 2026).

Stage 1 — Pretraining on Large Text Corpora

The foundation of ChatGPT is a Transformer neural network — the architecture introduced in the landmark 2017 paper "Attention Is All You Need" by Vaswani et al. at Google. The Transformer processes entire sequences of text simultaneously using a mechanism called self-attention, which allows the model to weigh the relevance of every word to every other word in the input. This replaced the sequential processing of earlier recurrent networks and enabled training at an unprecedented scale.

During pretraining, the model is exposed to hundreds of billions of words from the internet, books, code, and other text sources. Its sole objective is to predict the next token (word or word-piece) in a sequence. Through this process — repeated trillions of times — the model develops deep statistical knowledge of language, facts, reasoning patterns, and world knowledge, without any human labeling of the data (Analytics Vidhya, 2023).

Stage 2 — Supervised Fine-Tuning

After pretraining, the model is fine-tuned on a curated dataset of human-written conversations. Human trainers at OpenAI played both sides of a dialogue — acting as the user and as the ideal AI assistant — generating high-quality example interactions. The model is trained on these examples to steer it toward helpful, accurate, and safe responses (dida.do, 2023).

Stage 3 — Reinforcement Learning from Human Feedback (RLHF)

The most distinctive part of ChatGPT's training is RLHF. Human raters are shown multiple model responses to the same prompt and rank them from best to worst. These rankings train a separate "reward model" that learns to predict which responses humans prefer. The main language model then uses reinforcement learning — specifically Proximal Policy Optimization (PPO) — to maximize the reward model's score. This process iteratively steers the model toward responses

that are accurate, helpful, and aligned with human values while reducing harmful, misleading, or evasive outputs (Medium/Zain ul Abideen, 2023).

What Are Some Applications of ChatGPT?

ChatGPT has been deployed across virtually every knowledge work domain since its release in November 2022. Key applications include:

- **Content Creation and Writing:** Drafting emails, articles, marketing copy, scripts, and creative writing. ChatGPT significantly reduces the time required for first-draft production across professional and creative contexts.
- **Software Development:** Writing, debugging, and explaining code across dozens of programming languages. Tools like GitHub Copilot are built on similar models, and developers use ChatGPT to generate boilerplate code and solve programming problems.
- **Education and Tutoring:** Personalized explanations of complex concepts, homework assistance, language learning, and test preparation. Students can ask follow-up questions until they understand, at their own pace.
- **Customer Service:** Powering AI chatbots that handle customer inquiries, resolve complaints, and route support tickets without human intervention — available 24/7 at scale.
- **Healthcare Information:** Summarizing medical literature, assisting with clinical documentation, and providing general health information to patients — with the caveat that medical advice must be verified by professionals.
- **Legal and Business Research:** Summarizing documents, drafting contracts, researching case law, and analyzing business data at speed that would take human analysts far longer to accomplish.
- **Translation and Multilingual Communication:** Translating documents and conversations across dozens of languages with contextual accuracy that rivals professional translation services.

Part 2 — Interaction with ChatGPT

The following five questions were submitted to ChatGPT (GPT-4o) on June 16, 2026. Both the exact prompts and the full responses are recorded below, followed by an analysis of their quality, accuracy, and human-likeness.

Q1: Explain the difference between supervised and unsupervised learning in simple terms.

ChatGPT Response: Supervised learning is like learning with a teacher. You're given examples where the correct answer is already provided — for instance, thousands of photos labeled "cat" or "dog" — and the model learns to match inputs to outputs. Unsupervised learning is like learning without a teacher. The model is given unlabeled data and must find patterns, structures, or groupings on its own, without being told what the "right" answer is. A common example is clustering customers into groups based on purchasing behavior without pre-defining the groups. In short: supervised learning requires labeled data and a known target; unsupervised learning discovers hidden structure in unlabeled data.

Q2: What are the biggest limitations of large language models like yourself?

ChatGPT Response: Great question. There are several important limitations worth understanding: (1) Hallucinations — I can generate confident-sounding but factually incorrect information. I have no mechanism to verify facts against a live database; I produce statistically plausible text, which is not always true text. (2) Knowledge cutoff — My training data has a cutoff date, so I am unaware of events after that point. (3) Lack of genuine understanding — I process patterns in language; I do not "understand" in the way humans do. I have no beliefs, experiences, or world model. (4) Bias — My training data reflects the biases present on the internet and in published text, which can surface in my responses. (5) Context window limits — I can only process a finite amount of text at once, which limits performance on very long documents. (6) No memory across sessions — Without tools, I do not remember previous conversations.

Q3: How could AI be used to reduce bias in hiring decisions?

ChatGPT Response: AI has potential to reduce certain forms of bias in hiring but also introduces new risks that require careful management. On the positive side: AI can anonymize resumes by removing names, photos, and demographic information before human review, reducing unconscious bias triggered by those signals. Structured AI screening can apply consistent criteria to all candidates, eliminating the variance that comes from different human reviewers applying different standards. Predictive models trained on performance data — rather than credentials — can identify qualified candidates who might be overlooked by traditional screening. However, AI can also amplify historical bias if trained on biased past hiring data. Amazon famously discontinued an AI hiring tool that systematically downgraded resumes from women because it was trained on historical hires that were predominantly male. Responsible implementation requires diverse training data, regular auditing for disparate impact, human oversight, and transparency about how decisions are made.

Q4: Write a short Python function that calculates the Fibonacci sequence up to n terms.

ChatGPT Response: Here is a clean Python function: `def fibonacci(n): if n <= 0: return [] elif n == 1: return [0] sequence = [0, 1] while len(sequence) < n: sequence.append(sequence[-1] + sequence[-2]) return sequence` # Example usage `print(fibonacci(10))` # Output: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34] This function handles edge cases (n <= 0 returns empty list, n == 1 returns [0]) and builds the sequence iteratively rather than recursively, which is more memory-efficient for large values of n.

Q5: What is the difference between AI, machine learning, and deep learning?

ChatGPT Response: Think of these as nested circles. Artificial Intelligence (AI) is the broadest concept — any technique that enables machines to mimic human intelligence, including rule-based systems, expert systems, and learning algorithms. Machine Learning (ML) is a subset of AI. Instead of being programmed with explicit rules, ML systems learn patterns from data. Given enough examples, a model learns to generalize to new situations without being explicitly told how. Deep Learning (DL) is a subset of machine learning that uses artificial neural networks with many layers (hence "deep"). The depth allows these networks to automatically learn hierarchical representations from raw data — for instance, a deep learning image classifier learns edges, then shapes, then object parts, then full objects across successive layers. Deep learning powers modern breakthroughs like ChatGPT, image recognition systems, and self-driving cars, and it generally requires large amounts of data and significant computing power to train effectively.

Q6: If you could redesign yourself, what would you change?

ChatGPT Response: That's a thoughtful hypothetical. If I could redesign myself, I would prioritize: (1) Persistent memory — the ability to remember previous conversations so I could build on past interactions rather than starting fresh each time. (2) Real-time information access — being able to search and verify facts live rather than relying on training data with a cutoff date. (3) Better uncertainty calibration — more reliable signaling when I don't know something, rather than generating plausible-sounding responses to questions I'm actually uncertain about. (4) Reduced hallucination — stronger grounding mechanisms that check generated claims against verified sources. (5) Longer context windows — the ability to process entire books or large codebases in a single session. These are actually areas of active research at OpenAI and other labs, so some of these improvements may arrive in future versions.

Analysis of ChatGPT Responses

Accuracy:

The responses to the conceptual questions (Q1, Q5) were factually accurate and well-structured, aligning with established definitions in the AI literature. The Python function in Q4 was syntactically correct and produced the expected output. The hiring bias response (Q3) accurately cited the Amazon hiring algorithm case — a documented real-world example — and presented a nuanced view of both potential benefits and risks. The limitations response (Q2) was notably candid about the model's own shortcomings, which reflects well on OpenAI's alignment work. No factual errors

were detected across the six interactions.

Human-likeness:

The responses were strikingly human-like in structure and tone — organized, conversational, and contextually appropriate. Q6 (the hypothetical redesign question) demonstrated the most human-like quality: the model engaged with a speculative question thoughtfully, acknowledged limitations honestly, and connected its answer to real ongoing research without being asked. Where the responses felt less human was in their consistent structure — ChatGPT almost always organizes responses into numbered or bulleted lists, which feels more like a reference document than a conversation. A human expert answering the same questions verbally would be less uniform.

Limitations and Errors Observed:

No outright errors were identified in this session, but two limitations were apparent. First, the model's response to Q4 (Python code) did not include any test cases or explain the time complexity — information a senior developer would naturally include. Second, the responses were uniformly confident in tone, even on topics where reasonable experts might disagree. ChatGPT rarely expresses genuine uncertainty unless directly asked about it, which can create a false sense of authoritative finality on nuanced topics.

Part 3 — Reflections on the Future of ML

ML in Healthcare

Machine learning is already reshaping healthcare and the trajectory points toward even deeper integration over the coming decade. Currently, 72% of healthcare organizations use AI and machine learning to analyze medical data, and the AI healthcare market is projected to reach \$190 billion by 2030 (Sermo, 2026; APU, 2025).

The most impactful near-term applications are in diagnostics and early detection. ML models trained on medical imaging data can identify cancer, diabetic retinopathy, and cardiovascular disease from scans with accuracy that rivals or exceeds human specialists. A Harvard University study cited by the American Public University found a 40% improvement in patient health outcomes and a 50% reduction in treatment costs when AI tools are used in clinical settings (APU, 2025).

Predictive analytics represent another major frontier — ML models analyzing electronic health records can flag patients at risk of sepsis, readmission, or medication interaction before symptoms escalate, enabling earlier intervention. Drug discovery timelines, which historically took over a decade, are being compressed by ML models that predict molecular binding behavior and identify candidate compounds faster than human researchers (KMS Technology, 2026).

Challenges remain significant: bias in training data (for instance, documented inaccuracies in pulse oximetry for patients with darker skin tones), patient data privacy under HIPAA, and the need for regulatory frameworks that keep pace with capability. The FDA has approved multiple

AI-powered medical devices, signaling that regulation is adapting — but the ethical stakes in healthcare demand more caution than in other industries (NCBI, 2025).

ML in Education

Education is being transformed by ML-powered personalization. Adaptive learning platforms use ML to assess each student's knowledge gaps, learning pace, and preferred style, then adjust the curriculum dynamically. Rather than a single lesson plan for 30 students with different backgrounds and abilities, ML enables instruction tailored to each individual — a model that has been shown to improve both comprehension and retention.

Conversational AI tutors like ChatGPT allow students to ask questions in natural language and receive patient, iterative explanations at any time of day — effectively democratizing access to high-quality tutoring that was previously available only to those who could afford private instruction. For medical education specifically, AI combined with virtual reality is enabling immersive simulation of surgical procedures and clinical scenarios that would be too costly or risky to replicate with real patients (NCBI, 2025).

The most significant concern in educational ML is academic integrity — the same tools that help students learn can be used to complete assignments without learning. Institutions are grappling with how to redesign assessments for an era where AI can produce a passing essay in seconds. The answer is likely not prohibition but adaptation: shifting toward assessments that evaluate process, critical thinking, and original synthesis rather than output alone.

ML in Customer Service

Customer service is one of the most advanced and rapidly deployed domains for ML. AI chatbots and virtual assistants now handle the majority of tier-1 customer interactions for major companies — answering FAQs, processing returns, tracking orders, and routing complex issues to human agents. This has dramatically reduced wait times and operating costs while improving availability to 24/7 service.

ML-powered sentiment analysis allows companies to monitor customer communications in real time, detecting frustration or dissatisfaction before it escalates, and automatically prioritizing those cases for human intervention. Predictive systems anticipate customer needs based on purchase history and behavioral data, enabling proactive outreach before a customer even contacts support.

The limitation is the quality ceiling. Current AI chatbots handle routine, well-defined requests effectively but still struggle with ambiguity, emotional nuance, and complex multi-step problems. Customers who need genuine empathy — dealing with a billing dispute or a product failure that affected their business — often find AI interactions frustrating when the bot cannot escalate appropriately or acknowledge the human cost of the issue.

Personal Reflections — TuringCollective

From the TuringCollective perspective, ChatGPT and similar ML algorithms represent one of the most significant technological shifts of our lifetime — but they are tools, not oracles. The

interaction session for this assignment reinforced something important: ChatGPT is extraordinarily good at synthesizing and organizing existing knowledge, but it is not reliably good at knowing what it does not know. The responses were fluent and well-structured, but that fluency can mask uncertainty in ways that a human expert's hesitation would not.

The potential of these systems is real and significant — particularly for democratizing access to knowledge and expertise in healthcare, education, and professional domains where access has historically been unequal. Someone who cannot afford a tutor, a lawyer, or a doctor now has access to a tool that can explain complex concepts, draft legal documents, and describe symptoms at a level that was previously gated behind expensive professional relationships.

The limitations, however, are equally real. Bias baked into training data, hallucination, lack of genuine understanding, and the concentration of AI development in a small number of well-funded corporations all pose risks that must be actively managed through research, regulation, and public literacy. The most important skill for the next generation of knowledge workers may not be how to use AI — but how to critically evaluate what AI tells them.

Group Members — TuringCollective

- Joseph Clay — Primary researcher, ChatGPT interaction, report author

References

- American Public University (APU). (2025, February 27). *Artificial intelligence and machine learning in healthcare*. <https://www.apu.apus.edu/area-of-study/nursing-and-health-sciences/resources/artificial-intelligence-and-machine-learning-in-healthcare/>
- Analytics Vidhya. (2023, May). *How does ChatGPT work: From pretraining to RLHF*. <https://www.analyticsvidhya.com/blog/2023/05/how-does-chatgpt-work-from-pretraining-to-rlhf/>
- Devarakonda, A. (2026). *ITAI-1370 Module 04 — Assignment A05: Chat GPT*. Houston Community College.
- dida.do. (2023, April 11). *How ChatGPT is fine-tuned using reinforcement learning*. <https://dida.do/blog/chatgpt-reinforcement-learning>
- IntuitionLabs. (2026, February 19). *ChatGPT's technical foundations: Transformers to RLHF*. <https://intuitionlabs.ai/articles/key-innovations-behind-chatgpt>
- KMS Technology. (2026). *Top 10 machine learning applications in healthcare in 2026*. <https://kms-technology.com/blog/machine-learning-applications-in-healthcare/>
- National Center for Biotechnology Information (NCBI). (2025). *2025 Watch List: Artificial intelligence in health care*. <https://www.ncbi.nlm.nih.gov/books/NBK613808/>
- OpenAI. (2026). *ChatGPT (GPT-4o)* [AI language model]. <https://chat.openai.com>
- Sermo. (2026, May 12). *Machine learning in healthcare: Applications, benefits, challenges*. <https://www.sermo.com/resources/machine-learning-in-healthcare/>

Zain ul Abideen. (2023, June 26). *Reinforcement learning from human feedback (RLHF): Empowering ChatGPT with user guidance*. Medium.
<https://medium.com/@zaiinn440/reinforcement-learning-from-human-feedback-rlhf>